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Duration	2 mins 18 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Bubble Sort
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Bubble Sort
b. Linear Search
c. Binary Search
d. Merge Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort

c. Insertion sort

d. Bubble sort

The correct answer is:
Insertion sort

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470

b. 80,150,200,470,40,90,300,400,70,120

c. 200,470,80,150,40,90,300,400,70,120

d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists

b. It recursively sorts subproblems

c. It bubbles up the largest element in each pass

d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **6**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Comparing each element with the others
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **7**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It makes the data unsorted
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting makes it harder to search for items
- c. Sorting decreases the efficiency of selection operations
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Bubble Sort
c. Quick Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort is less efficient for large lists
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **11**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is based on the divide-and-conquer approach
c. It does not require recursion
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **12**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Bubbling up the smallest element
- c. Dividing the list into halves
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Merging two datasets
- c. Arranging data in a specific order
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **14**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Selection
- b. Searching
- c. All of the above
- d. Duplicates

The correct answer is: All of the above

Question **15**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Searching for data in a list
b. Arranging data in a particular format
c. Inserting data into a list
d. Deleting data from a list

The correct answer is: Arranging data in a particular format